

Project Design Phase-II

Solution Requirements (Functional & Non-functional)

Date	30 July 2026
Team ID	Individual / Solo Project
Project Name	Automated Network Request Management in ServiceNow
Student	Bandhanadam Antony
Maximum Marks	4 Marks

Functional Requirements

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	Network Service Discovery	• Browse the Service Catalog for approved network access forms (e.g., VPN, Firewall, Port forwarding) • Search for network services by name or category • View security risk tier and approval requirement for each item
FR-2	Request Submission	• Submit a request with business justification, source IP, and destination IP/Port • Auto-populate employee and target device details from the CMDB • Receive confirmation once the request is submitted
FR-3	Approval Workflow	• Route the request to the employee's manager for initial approval • Route high-risk requests to the InfoSec team for secondary review • Approve or reject a request directly within ServiceNow • Escalate or delegate approval automatically after a set timeout
FR-4	Risk & Policy Compliance	• Check requested configuration against baseline security policies before fulfillment • Block the request and notify InfoSec when blacklisted ports/protocols are requested • Track active firewall policies for continuous compliance audits
FR-5	Fulfillment & Configuration	• Queue approved, compliant requests for network fulfillment • Automate remote configuration for standard pre-approved network changes via Ansible/IntegrationHub • Mark the request as fulfilled and update the ticket status upon success
FR-6	Notifications	• Send email/portal confirmation on submission, approval, and completion • Send reminder notifications to managers for pending approvals
FR-7	Reporting & CMDB Tracking	• Log configured network changes and update CI relationships in the asset inventory (CMDB) • Provide a dashboard of network request volumes and security compliance status • Send audit reminders for temporary network access configurations

Non-functional Requirements

Following are the non-functional requirements of the proposed solution.

NFR No.	Non-Functional Requirement	Description
NFR-1	Usability	The Service Catalog item and request form should be simple enough for any employee to submit a network request without training or network engineering knowledge.
NFR-2	Security	Access to the catalog item and approval actions should be restricted by role, and all requests, approvals, and configuration changes should be logged for security audits.
NFR-3	Reliability	The approval routing, CMDB checks, and fulfillment workflow should run consistently without dropped or duplicated requests to prevent network outages.
NFR-4	Performance	Catalog search and request submission should respond within a few seconds even during peak usage periods.
NFR-5	Availability	The Service Catalog and request workflow should be available during standard business hours across all supported regions, with minimal planned downtime.
NFR-6	Scalability	The workflow should support a growing number of employees, endpoints, network devices, and concurrent requests without degrading performance.